Course Code	Course Title	Credits
CSE 2200	Viva-Voce	1.0
CSE 2201	Design and Analysis of Algorithms	3.0
CSE 2202	Design and Analysis of Algorithms Sessional	1.0
CSE 2203	Database Management System	3.0
CSE 2204	Database Management System Sessional	1.0
CSE 2205	Advanced Concepts of Programming Languages	3.0
CSE 2206	Advanced Concepts of Programming Languages Sessional	1.0
CSE 2207	Hardware/Software Project	1.0
CSE 2209	Operating Systems	3.0
CSE 2210	Operating Systems Sessional	1.0
MATH 2211	Numerical Methods and Analysis	3.0
	Total	**21.0**

Course Code: CSE 2200 **Course Title:** Viva-Voce **Credits:** 1.0

Measure and evaluate the students of Computer Science and Engineering (CSE) Department through oral examination based on their previous taught/learned courses.

Course Code: CSE 2201 **Course Title:** Design and Analysis of Algorithms **Credits:** 3.0

Introduction to Algorithm: What Kind of Problems are solved by Algorithms? Algorithms as a Technology, Analyzing Algorithms, Searching, Linear Search, Binary Search, Sorting, Insertion Sort, Selection Sort, Bubble Sort, Algorithmic paradigms

Divide and Conquer: Property, Recurrence, Power of an element, Merge Sort, Counting Inversion in Merge Sort, Quick Sort, Randomized Quick Sort

Sorting: Heap Sort, Linear Sorting Algorithms, Counting Sort, Radix Sort

Greedy Algorithms: Property, Problems, Task Scheduling, Fractional Knapsack, Greedy Coin Change, Huffman Coding

Dynamic Programming: Property, Problems: Fibonacci Number, Rod Cutting, Knapsack, Coin Change, Hill Climbing, LCS, Bit-Masking

Graph Algorithms: Graph Representation, BFS, DFS, Topological Sort, Shortest Path, Single Source Shortest Path: Dijkstra Algorithm, Bellman Ford Algorithm, All Pair Shortest Path: Floyd Warshall, Johnsons Algorithm, Minimum Spanning Tree: Prims Algorithm, Kruskal Algorithm.

Network Flow and Matching: SCC, Articulation Point and Bridge using (Tarjan algorithm), Flow networks, Ford-Fulkerson method, Max Flow Min Cut Problem, Dinic's Algorithm, Maximum Bipartite Matching, Stable Marriage Problem, Weighted Bipartite Matching, Min Cost Max Flow, The naïve string matching algorithm, String Matching with Finite Automata, The Knuth-Morris-Pratt algorithm; The naïve string matching algorithm, String Matching with Finite Automata, The Knuth-Morris-Pratt algorithm

Books Recommended:

1. Goodman and Hedetniemi, "Introduction to Design and Analysis of Algorithm".

2. Horawitz E., Sahni S., Rajasekaram S., Computer Algorithms, W.H. Freeman & Co.
3. Cormen T.H., Leiserson C.E., Ribest R.L., Introduction to Algorithms, MIT Press.
4. E. Horowitz and S. Sahni, "Computer Algorithms", Galgotia.

Course Code: CSE 2202	**Course Title:** Design and Analysis of Algorithms Sessional	**Credits: 1.0**

In addition to the theoretical underpinnings presented in CSE 2201, this lab course is available. Students will be able to practice and apply academic concepts in a real-world situation through practical exercises. In addition to improving their knowledge, the interactive lab exercises will assist students in gaining the technical abilities necessary for their success in the workplace.

Books Recommended:

1. Goodman and Hedetniemi, "Introduction to Design and Analysis of Algorithm".
2. Horawitz E., Sahni S., Rajasekaram S., Computer Algorithms, W.H. Freeman & Co.
3. Cormen T.H., Leiserson C.E., Ribest R.L., Introduction to Algorithms, MIT Press.
4. E. Horowitz and S. Sahni, "Computer Algorithms", Galgotia.

Course Code: CSE 2203	**Course Title:** Database Management System	**Credits:** 3.0

Basic Concepts: Review of evolution of data management, file management vs. data management; characteristics of database approach; major roles involved in a database environment, database architecture - components of a DBMS; database features, advantages and disadvantages, data abstraction and data models, database languages, transaction management, storage management, database administrator, database users.

Relational database: Basic concept, key/mapping constraints, relational algebra; Relational commercial language: SQL, QUEL; Integrity constraints: Domain Constraints, Referential integrity, Functional Dependencies; Relational database design, Storage and file structure: File Organization, Organization of records in files.

E-R Model: Basic concept, Design Issues, Mapping Constraints, Key conception, Strong and Week Entity set.

Distributed database: Introduction to Visual .NET/ C# in Visual Studio, Coding with Visual .NET/ C# Data types (built in data types and user defined data types), Control Structure, objects, class, module, Project and functions (Private, Public) etc.

Client-Server Systems: Review of client server computing, application partitioning, 2-layer and 3-layer architectures; middleware concepts, use of APIs in client-server components, GUI features; middleware – RPC, message-oriented – middleware; distributed computing environment (DCE) components, DCE RPC, DCE services, object request brokers; Common Object Request Broker Architecture (CORBA), distributed object services.

Study of a Commercial RDBMS: Product architecture, product components; run-time architecture; data definitions, system catalog, SQL implementation, embedded SQL implementation (static and dynamic), API implementation, object-relational features, integrity support, security support, database loading; database backup/restore; Codd's 12 rules as applied to the product; major strengths of the product; major limitations of the product, interfaces to client development environments and the features of such environment(s).

Books Recommended:

1. Silverschatz A., Korth H., Sudarshon S., "Database System Concepts".
2. Elmasri R., Navathe S.B., "Fundamentals of Database Systems", Addison-Wesley.

Course Code: CSE 2204	**Course Title:** Database Management System Sessional	**Credits: 1.0**

In addition to the theoretical underpinnings presented in CSE 2203, this lab course is available. Students will be able to practice and apply academic concepts in a real-world situation through practical exercises. In addition to improving their knowledge, the interactive lab exercises will assist students in gaining the technical abilities necessary for their success in the workplace.

Books Recommended:

1. Silverschatz A., Korth H., Sudarshon S., "Database System Concepts".
2. Elmasri R., Navathe S.B., "Fundamentals of Database Systems", Addison-Wesley.

Course Code: CSE 2205	**Course Title:** Advanced Concepts of Programming Languages	**Credits:** 3.0

Fundamentals of Python Programing: The Concept of Data Types, Variables, Assignments, Immutable Variables, Numerical Types, Arithmetic Operators and Expressions, Comments in the Program, Understanding Error Messages.
Basic Algorithm Implementations using Python: Lists, Tuples, and Dictionaries, Basic List Operators, Replacing, Inserting, Removing an Element; Searching and Sorting Lists, Dictionary Literals, Adding and Removing Keys, Accessing and Replacing Values, Traversing Dictionaries.
Design Functions with Python: Hiding Redundancy, Complexity, Arguments and Return Values, Formal Vs Actual Arguments, Named Arguments, Program Structure and Design, Recursive Functions.
Classes and OOP in Python: Classes, Objects, Attributes and Methods, Defining Classes, Design with Classes, Data Modeling, Persistent Storage of Objects, Inheritance, Polymorphism, Operator Overloading (Eq, Str, Etc), Abstract Classes, Exception Handling, Try Block.
HTML: Understanding the structure of HTML documents, Basic tags: <html>, <head>, <title>, <body>, Creating and organizing content with headings, paragraphs, lists, and links, Images, audio, and video embedding, Tables and forms, Semantic HTML and best practices
CSS: Basics of CSS: Syntax, selectors, and properties, Inline, internal, and external CSS, Understanding the box model, Layouts: Flexbox, Grid, and positioning, Styling text and fonts, Responsive design and media queries, Animations and transitions
JavaScript: Basic syntax and data types, Variables, operators, and expressions, Functions and control flow, Selecting and manipulating DOM elements, Event handling and listeners, Form validation, Asynchronous JavaScript: Callbacks, Promises, and Async/Await, JavaScript modules and ES6 features, Working with APIs and AJAX
PHP: Basics of PHP syntax and variables, Control structures and functions, working with forms and user input, Introduction to MySQL, connecting to a database using PHP, Performing CRUD operations, Sessions and cookies, File handling and uploads, Security practices in PHP

ASP.NET: Overview of the ASP.NET framework, setting up the development environment, Creating and running a simple ASP.NET application, Understanding Web Forms and MVC architecture, Working with Razor syntax, Building dynamic web pages and forms

Books Recommended:

1. Barry P. and Griffiths D., "Head First Programming: A Learner's Guide to Programming Using the Python Language", O'Reilly Media.
2. Sweigart A., "Automate the boring stuff with Python: practical programming for total beginners".
3. Rob Birdwell, Matt Butter Ollie Cornes," ASP.NET with C#.NET".
4. MSDN Training Module.
5. Web Development with Blazor: A practical guide to start building interactive UIs with C# 11 and .NET 7.

Course Code: CSE 2206 **Course Title:** Advanced Concepts of Programming Languages Sessional **Credits:** 1.0

This lab course complements the theory course CSE 2205 by providing hands-on exercises to solidify theoretical concepts through practical application and the students will develop essential skills through interactive lab activities.

Books Recommended:

1. Barry P. and Griffiths D., "Head First Programming: A Learner's Guide to Programming Using the Python Language", O'Reilly Media.
2. Sweigart A., "Automate the boring stuff with Python: practical programming for total beginners".
3. Rob Birdwell, Matt Butter Ollie Cornes," ASP.NET with C#.NET".
4. MSDN Training Module.
5. Web Development with Blazor: A practical guide to start building interactive UIs with C# 11 and .NET 7.

Course Code: CSE 2207 **Course Title:** Hardware/Software Project **Credits:** 1.0

The objective of the project work is to develop software or hardware systems of applied nature encompassing database, networks, multimedia, web design, electronic commerce, object-oriented programming, software development etc. A teacher shall supervise the project work. The project has to be completed and evaluated in 1.0 credit.

Course Code: MATH 2211 **Course Title:** Numerical Methods and Analysis **Credits:** 3.0

Overview of Numerical Analysis: Introduction to Numerical Analysis, Characteristics of Numerical Analysis, Importance of Numerical Analysis, Application of Numerical Analysis in Computer science.

Solution of Equations: Bisection Method, Regular-Falsi Method, Graphical Method, Secant Method, Newton-Raphson Method.

Interpolation: Forward Differences, Backward Differences and Divided Differences, Newton's Forward Differences Interpolation Formula, Newton's Backward Differences Interpolation Formula, Divided Differences Interpolation Formula, Lagrange's Interpolation Formula.

Errors: Introduce with Round-off errors, Types of Errors, Floating Point and Convergence.
Elimination Methods: Gauss Elimination Method, Jordan Elimination Method.
Numerical Integration: Trapezoidal Rule**,** Simpson's 1/3 and 3/8 Rules, Weddell's Rules.

Books Recommended:

1. Terrence J. Akai, "Applied Numerical Methods for Engineers", Wiley.
2. Richard L. Burden, J. Douglas Faires, "Numerical Analysis".
3. J. Stoer, R. Bulirsch, R. Bartels and W. Gautschi, "Introduction to Numerical Analysis", Springer, 3rd Edition, December 1, 2018.

Course Code: CSE 2209 **Course Title:** Operating Systems **Credits:** 3.0

Operating System Overview: OS functions, evolution of OS functions, batch processing systems, single user multiprogramming systems, time sharing systems, real-time operating systems, OS structure.
Processes: Process definition, process control, interacting processes, implementation of interacting processes, threads.
Scheduling: Scheduling policies, job scheduling, process scheduling, process management in UNIX, scheduling in multiprocessor OS, queuing in scheduling, scheduling algorithms, priority control, introduction to multiprocessor and real-time scheduling.
Deadlocks: Definitions, resource status modeling, handling deadlocks, deadlock detection and resolution, deadlock avoidance, mixed approach to deadlock handling.
Process synchronization: Implementing control synchronization, critical sections, classical process synchronization problems, evolution of language features for process synchronization, semaphores, critical regions, conditional critical regions, monitors, concurrency.
Memory Management: Principles, requirements and design of memory management system, program loading and linking. Virtual Memory: locality, page table, translation lookaside buffer, segmentation, software implementation, load control.
I/O Management and Disk Scheduling: Organization of the I/O function. Direct memory access. Design issues. I/O buffering. Disk I/O. Disk cache. Example systems.
File Management: Overview, File management systems. File organization and access, file directories. Sharing of files. Record blocking. Secondary storage management, file systems protection and security; design and implementation methodology, Example systems.

Books Recommended:

1. Tanenbaum A.S., Modern Operating Systems, Prentice Hall.
2. Silberschatz A., Galvin P.B., Operating System Concepts, John Wiley & Sons.
3. B. Hausen, "Operating System Principles", Prentice-Hall of India.

Course Code: CSE 2210 **Course Title:** Operating Systems Sessional **Credits:** 1.0

In addition to the theoretical underpinnings presented in CSE 2209, this lab course is available. Students will be able to practice and apply academic concepts in a real-world situation through practical exercises. In addition to improving their knowledge, the interactive lab exercises will assist students in gaining the technical abilities necessary for their success in the workplace.

Books Recommended:

1. Tanenbaum A.S., Modern Operating Systems, Prentice Hall.
2. Silberschatz A., Galvin P.B., Operating System Concepts, John Wiley & Sons.
3. B. Hausen, “Operating System Principles”, Prentice-Hall of India.